Quark Scan: Fitting vs. Sampling, and Why the Envelope M_{KK} Bound Was So Low

May 26, 2026

1 What the headline outcome is

In response to the collaborator's critique, we did two specific things:

1. **Promoted the quark mass targets to PDG-faithful $\overline{\mathrm{MS}}$ values.** Previously the scan was matching to ad-hoc numbers nominally evaluated at the KK scale (~ 3 TeV); the strange quark target alone was off by $\sim 50\%$. We replaced these with the PDG 2024 $\overline{\mathrm{MS}}$ masses and ran them down to a common reference scale of $\mu = m_t = 163.5$ GeV using four-loop QCD mass running with the standard $n_f = 6 \to 5 \to 4$ threshold matching at the heavy-quark thresholds.

2. **Re-ran the dense scan** ($100\,\mathrm{k}$ points, $r \in [0.05, 1.0]$, $M_{\mathrm{KK}} \in [1, 20]$ TeV) under the corrected targets and a tightened 2σ PDG tolerance on both quark masses and the four CKM observables $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$, J.

The publication-style M_{KK} exclusion plot from this re-run is *visually almost identical* to the previous version. The lowest allowed M_{KK} stayed at ~ 1.2 TeV in both the old and the corrected scans. The collaborator was right that the targets had been wrong, and we fixed them, and the headline M_{KK} number did not move.

The rest of this note explains why that is correct, and what we should be plotting instead.

Reading guide. §2–§4 introduce the physics needed to read the figures: what "new physics" means, what the FCNC "ratio" is, and the gauge-coupling convention that sets the absolute M_{KK} scale. §5–§6 explain why the envelope plot has been answering an existence rather than a typicality question, and what the optimiser's Yukawas actually look like. §7–§9 present the alternative "RS-anarchy" analysis, including where the priors come from and how a random draw can satisfy PDG masses without any fitting. §10–§11 give the resulting M_{KK} bound and its sensitivity to the gate definition. §12 reports five follow-up scans that probe how the bound responds to statistics, the c-pattern, the Y-prior, the PDG-gate tightness, and the CFW comparison. Appendix A.3 collects the scope and approximation caveats for the KK tower, EWPO floor, gauge-coupling convention, and lepton sector.

2 What "new physics" is in this scan

The 5D Randall–Sundrum model has, in addition to the Standard Model fields, an infinite tower of Kaluza–Klein (KK) excitations of every gauge boson and fermion. The lightest gluon excitation is the "KK-gluon"; its mass M_{KK} is the headline number we are trying to constrain. The scan below keeps this lightest colour-octet mode only; the higher-mode tower truncation is treated in Appendix A.3.

The KK-gluon is a colour-octet vector boson whose 5D wave function peaks at the IR brane. Crucially, its couplings to the SM quarks are *flavour-non-universal*: a quark that is itself IR-localised (like the top) couples strongly, while a UV-localised quark (like the up) couples weakly. This non-universality is exactly what gives the model its attractive flavour-hierarchy explanation, but it is also what generates new flavour-changing neutral currents.

When one integrates out the lightest KK-gluon, the resulting effective four-quark operators contribute to neutral meson mixing in five places:

$$\varepsilon_K, \quad \Delta m_K, \quad \Delta m_{B_d}, \quad \Delta m_{B_s}, \quad \Delta m_{D^0}. \tag{1}$$

These are the five $\Delta F = 2$ observables used throughout the scan. The first (ε_K) measures CP-violation in the kaon system; the next three measure the magnitudes of $K\bar{K}$, $B_d\bar{B}_d$, $B_s\bar{B}_s$ mixing; the last measures $D^0\bar{D}^0$ mixing.

In each of these five processes, the quantity that the model predicts is the new-physics contribution to the meson mixing matrix element, M_{12}^{NP}. This is what we mean by "new physics": the KK-gluon-induced piece of the mixing amplitude, on top of the Standard-Model piece M_{12}^{SM}. Both pieces are functions of the underlying 5D Yukawa matrices Y_u, Y_d, the bulk-mass parameters $c_{Q_i}, c_{u_i}, c_{d_i}$ that control where each fermion sits in the extra dimension, and M_{KK} itself.

3 What the FCNC "ratio" is, and what $\mathrm{ratio} = 1$ means

For each of the five mesons, the code in `quarkConstraints/deltaf2.py` computes a dimensionless number called `ratio_to_bound`. The definition is essentially the same in all five cases, but the K system deserves separate treatment because of CP violation:

- For $\Delta m_K, \Delta m_{B_d}, \Delta m_{B_s}, \Delta m_{D^0}$:
$$\mathrm{ratio} \;=\; \frac{\left|M_{12}^{\mathrm{NP}}\right|}{\left|M_{12}^{\mathrm{exp}}\right|}, \tag{2}$$
where $|M_{12}^{\mathrm{exp}}|$ is the experimentally measured mass-splitting amplitude. $\mathrm{ratio} = 1$ means the new-physics contribution alone equals the entire measured mixing amplitude.

- For ε_K (which is CP-violating and depends only on the *imaginary* part of M_{12}),
$$\mathrm{ratio} \;=\; \frac{\left|\,\varepsilon_K^{\mathrm{NP}}\,\right|}{\left|\,\varepsilon_K^{\mathrm{exp}} - \varepsilon_K^{\mathrm{SM}}\,\right|}, \tag{3}$$
with $|\varepsilon_K^{\mathrm{exp}} - \varepsilon_K^{\mathrm{SM}}| \approx 6.7 \times 10^{-5}$ in the audited constants (`deltaf2.py:725--729`). $\varepsilon_K^{\mathrm{NP}}$ is built from $\mathrm{Im}\, M_{12}^{\mathrm{NP}}$, not its magnitude. Here $\mathrm{ratio} = 1$ means the new physics fills the entire "room" between the SM prediction and the measurement, i.e. NP saturates the experimental–theoretical gap.

A draw is said to "pass all FCNC constraints" when $\max_{\text{five systems}}$ ratio < 1. This is one specific choice of gate, and we will examine its sensitivity in §11. For now the important thing is that ratio $= 1$ is the boundary at which the new-physics contribution is as large as the experiment allows.

4 What $g_s^\star$ is, and why the literature uses the values 1, 3, and 6

The KK-gluon couples to two SM quarks through QCD, so every entry of every ratio above is proportional to the square of an effective gauge coupling. Three different values of that coupling appear in different parts of the literature: $g_s \sim 1$, $g_s^\star \sim 3$, and $g_s^\star \sim 6$. These are not contradictions; they are the same physics described in three normalisation conventions, and which one a paper uses changes the M_{KK} bound by up to a factor of six. So this convention deserves its own section.

4D vs. 5D gauge coupling. Pure 4D QCD has a single coupling $g_s^{(4)}$, which runs with scale and takes the value $g_s^{(4)} \approx 1.05$ at $\mu = 3$ TeV (PDG running from $\alpha_s(M_Z) = 0.1180$). The 5D Randall–Sundrum gauge theory has an extra coupling: the bulk gauge coupling $g_s^{(5)}$ in the 5D Lagrangian $-\frac{1}{4(g_s^{(5)})^2} F_{MN}F^{MN}$, which has dimensions of $[\text{length}]^{1/2}$. After Kaluza–Klein reduction, the dimensionful $g_s^{(5)}$ and the dimensionless $g_s^{(4)}$ are related by

$$\frac{1}{(g_s^{(4)})^2} = \frac{\pi r_c}{(g_s^{(5)})^2}, \qquad \pi r_c k \equiv \log(M_{\text{Pl}}/\Lambda_{\text{IR}}) \approx 35 \quad \text{(the warp log)}. \tag{4}$$

The combination that controls strong-coupling effects in the 5D theory is the dimensionless number

$$g_s^\star \equiv g_s^{(5)}\sqrt{k} = g_s^{(4)}\sqrt{\pi r_c k} \approx g_s^{(4)} \cdot \sqrt{35} \approx 5.9\, g_s^{(4)}. \tag{5}$$

This $g_s^\star$ is the natural dimensionless coupling of the 5D gauge theory at the scale k. With $g_s^{(4)} \approx 1.05$ at $M_{\text{KK}} \sim 3$ TeV, equation (5) gives $g_s^\star \approx 6$. So the "6" is just $g_s^{(4)}$ multiplied by the warp-factor enhancement $\sqrt{\pi r_c k}$.

Why "3". The KK-gluon does not see the entire bulk uniformly. Its profile is peaked at the IR brane, which means its overlap with an IR-localised fermion is *not* the full $\sqrt{\pi r_c k}$ enhancement; the integral cuts off once the KK profile dies away. A careful evaluation of the KK-gluon–IR-fermion vertex (see e.g. Agashe *et al.* hep-ph/0606293, Csaki–Falkowski–Weiler 0804.1954) gives, for a fermion with $c < 1/2$,

$$g^{\text{eff}}_{\text{KK-gluon}\,ff} \approx g_s^{(4)}\, R(c), \qquad R(c = 1/2) \approx \sqrt{\pi r_c k}/\sqrt{2} \approx 4, \qquad R(c \to -\infty) \approx \sqrt{2\pi r_c k} \approx 8. \tag{6}$$

For UV-localised fermions ($c > 1/2$) the overlap factor is suppressed, $R(c \sim 0.6) \approx -0.2$. So depending on which c you anchor on, the "effective $g_s^\star$" that enters Wilson coefficients comes out to ~ 3 or ~ 6. The conventional "$g_s^\star = 3$" is what you get if you anchor on the canonical value

$$g_s^\star \approx \frac{g_s^{(4)}\sqrt{\pi r_c k}}{2} \approx 3, \tag{7}$$

which is also approximately the NDA upper bound on the dimensionless 5D coupling before the gauge sector becomes strongly coupled, $g_s^\star \lesssim 4\pi/\sqrt{N_{KK}\log(M_{\text{Pl}}/M_{\text{KK}})} \approx 3$. So "3" is a useful common comparison convention. Csaki–Falkowski–Weiler quote both sides of this ambiguity: their published 21 TeV RS marker is tied to a boundary-term convention with $g_s^\star \sim 6$, while their no-UV-boundary-term variant lowers $g_s^\star$ to ~ 3 and the same bound to 10.5 TeV.

So which is right? None of them is more right than another — they are different quantities. What matters is which coupling the Wilson-coefficient formula uses, and not double-counting:

- In our code (`quarkConstraints/couplings.py:101--124`), the Wilson coefficients C_1, C_4, C_5 are computed with the *perturbative* 4D coupling $g_s^{(4)} \approx 1.05$, and the $f_{\mathrm{IR}}(c)$ overlap factors carry the warp enhancement explicitly. The user can override this by passing `g_s_star = 3.0`, in which case the prefactor is replaced wholesale.
- The Csaki–Falkowski–Weiler formula puts the warp enhancement *into* $g_s^\star$ and writes the Wilson coefficient as $\propto (g_s^\star)^2/M_{\mathrm{KK}}^2$. Depending on UV boundary kinetic terms, they discuss both $g_s^\star \sim 6$ and $g_s^\star \sim 3$.
- The "$g_s^\star \sim 6$" value is the full warp-enhanced coupling $g_s^{(4)}\sqrt{\pi r_c k}$ in their tree-level matching convention. It is roughly twice the $g_s^\star = 3$ comparison convention used for the post-audit plots in this note.

The practical consequence is that the M_{KK} bound is defined up to a multiplicative factor that depends on how the gauge coupling enters the ratio formula. Since ratio $\propto g^2/M_{\mathrm{KK}}^2$, switching from $g_s^{(4)} \approx 1.05$ to $g_s^\star = 3$ multiplies every ratio by $(3/1.05)^2 \approx 8.2$, so a fixed acceptance threshold is reached at a M_{KK} value larger by a factor of $\sqrt{8.2} \approx 2.9$. In this paper the headline convention is $g_s^\star = 3$; perturbative $g_s \simeq 1.05$ numbers are reported as the code-default cross-reference. The $g_s^\star \simeq 6$ convention is used only for the CFW boundary-term comparison, as summarized in Appendix A.3.

5 Why the envelope plot is an existence statement

Now we can describe what the existing M_{KK} exclusion plot is actually computing. The scan iterates over a grid in $(r,\, M_{\mathrm{KK}})$. The parameter $r = \gamma_Q/\beta_Q$ is the ratio of two coefficients in the "Minimal-Flavor-Violation spurion"

$$C_Q \;=\; \alpha_Q \mathbb{1}_{3\times 3} + \beta_Q\, Y_d Y_d^\dagger + \gamma_Q\, Y_u Y_u^\dagger, \tag{8}$$

which determines the bulk mass parameters of the left-handed quark doublets in terms of the Yukawa matrices. (In short: r tunes how much the left-handed bulk profiles know about the up vs. down Yukawa structure.)

At each grid cell the scan calls a Levenberg–Marquardt least-squares solver, `fit_quark_sector`, which adjusts *simultaneously*:

- the rescaled Yukawa eigenvalues $\bar{y}_{u,i}, \bar{y}_{d,i}$ (six numbers),
- the spurion coefficients $\alpha_Q, \beta_Q, \gamma_Q$ and their right-handed analogues, which determine all nine $c_{Q_i}, c_{u_i}, c_{d_i}$,
- an SVD orientation matrix that fixes the relative basis of Y_u and Y_d (and hence the CKM matrix it produces),

to *minimise* a residual function with three groups of terms:

$$\chi^2_{\text{fit}} = \sum_{i=1}^{6}\Big(\frac{\log m_i^{\text{fit}} - \log m_i^{\text{PDG}}}{\sigma_{\log m_i}}\Big)^2 + \sum_{\alpha=1}^{4}\Big(\frac{V_\alpha^{\text{fit}} - V_\alpha^{\text{PDG}}}{\sigma_{V_\alpha}}\Big)^2 + \lambda_{\text{FCNC}}\sum_X [\text{ratio}_X]^2. \tag{9}$$

A grid cell is declared "allowed" if *any* optimum lands within 2σ on all six masses and four CKM entries, and below unity on all five FCNC ratios. So the published exclusion plot is answering this question:

> *Does there exist a choice of Yukawa eigenvalues, spurion coefficients, and orientation that simultaneously matches every PDG observable to 2σ and stays below the FCNC bounds?*

The answer turns out to be "yes, often, even at $M_{\text{KK}} \approx 1$ TeV". This is not because the model is fine, but because we gave the optimiser *eighteen continuous knobs* to play with at every grid cell.

6 What the envelope's Yukawas look like, and what is fine-tuned

To see what the optimiser is actually doing, we tabulate the $|Y_{f,ij}|$ of every entry of Y_u and Y_d over all $\sim 84\,000$ accepted points and split the histogram into diagonal and off-diagonal entries. Figure 1 compares this to a generic $\mathcal{O}(1)$ anarchic prior with $\text{Re}\, Y_{ij}, \text{Im}\, Y_{ij} \sim \mathcal{U}(-1.5, 1.5)$.

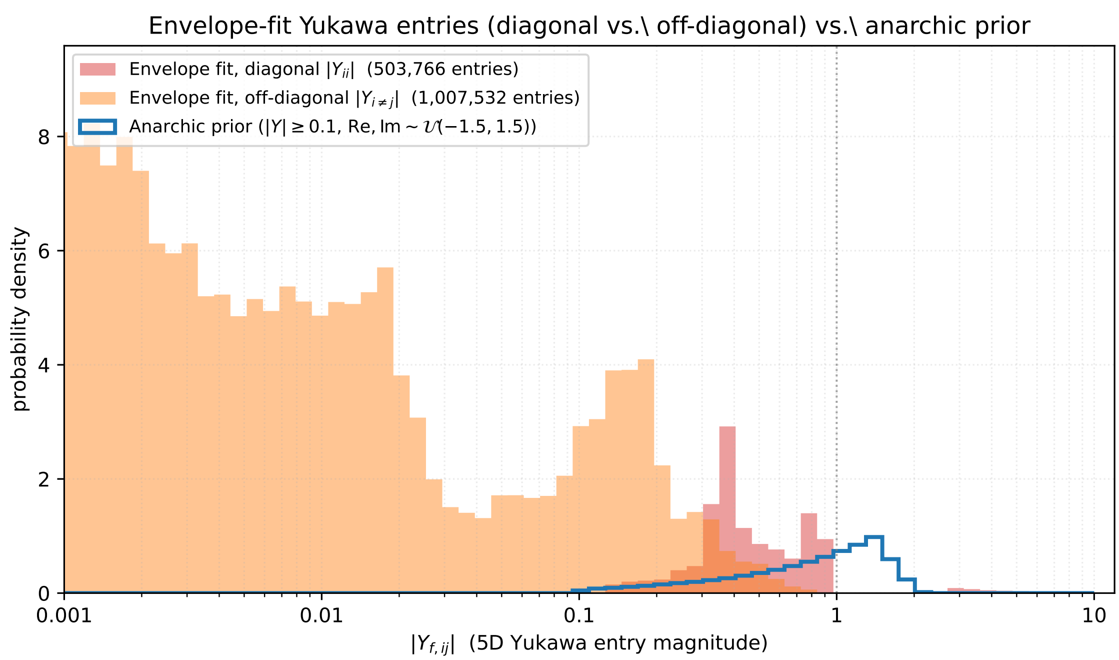

Figure 1: Distribution of fitted Yukawa-element magnitudes across the envelope-fit accepted points, split into diagonal entries $|Y_{ii}|$ (red) and off-diagonal entries $|Y_{i\neq j}|$ (orange), compared against the size distribution from a generic anarchic prior (blue line). The diagonal entries land where the SM masses require them: $|Y_{ii}|$ peaks broadly at ~ 0.5, with the top-Yukawa entry near $|Y| \sim 5$. The off-diagonal entries are pushed to small values by the optimiser, with a peak below $|Y| \sim 10^{-3}$. This is the signature of basis-aligned (low-FCNC) flavour structure.

The numbers are: diagonal entries have median $|Y_{ii}| \sim 0.7$ and 95th percentile ~ 4.5 (with $|Y_{u,33}| \sim 5$ for the top); these are properly $\mathcal{O}(1)$ and exactly what is needed to produce the SM mass eigenvalues given the envelope's $f_{\rm IR}$-factors. The off-diagonal entries have median ~ 0.17 but a long tail down to $|Y| \sim 10^{-3}$, with 5th percentile ~ 0.009. Roughly 30% of the off-diagonal entries are at $|Y| \leq 0.05$, a regime that an anarchic prior almost never visits.

What is being fine-tuned, and is it a "cancellation"? It is not a wild numerical cancellation between two separately large contributions. The optimiser is performing a basis alignment: it chooses the relative orientation of Y_u and Y_d (and of the bulk profiles, via the spurion ansatz (8)) so that the off- diagonal entries of $Y_u Y_u^\dagger$ and $Y_d Y_d^\dagger$ that drive the KK-gluon FCNC operators are simultaneously small. This is the same mechanism as Minimal Flavour Violation in 4D models: the flavour- violating operators are forced to be proportional to a single Yukawa spurion, which makes them small in the mass eigenbasis.

So the fine-tuning is not in any single parameter but in the *geometry* of the Yukawa matrices in flavour space. Concretely, the optimiser holds the three diagonal entries to their SM-required values and adjusts six off-diagonal entries (per matrix) and an SVD orientation to minimise a sum of FCNC penalties. This corresponds to choosing about 10 continuous numbers to lie in a small subspace of the 18-dimensional Yukawa space. The fitted accepted region is thin in this 18-dimensional sense, even though no individual entry is wildly fine-tuned.

A useful single-number summary is the median $|Y_{i\neq j}|/|Y_{ii}|$ ratio in the envelope fit: it sits at ~ 0.25, while the same quantity in the anarchic prior is ~ 1. A factor-four suppression of off-diagonals relative to diagonals is the quantitative signature of the alignment.

7 What we should plot instead: a measure statement

The phenomenologically meaningful question is the inversion of the existence question:

> *If the 5D Yukawa entries are drawn from an $\mathcal{O}(1)$ anarchic prior (as the model assumes), what fraction of draws satisfies the PDG and FCNC constraints at this $M_{\rm KK}$? Equivalently: above what $M_{\rm KK}$ does a typical anarchic point pass?*

This is the framing of the original RS-anarchy literature (Agashe–Contino–Pomarol–Sundrum hep-ph/0408134, Agashe–Perez hep-ph/0606293, Csaki–Falkowski–Weiler 0804.1954, and many others): the warped geometry is *supposed* to predict the SM mass and CKM hierarchies from anarchic 5D Yukawas via the IR-overlap factor

$$f_{\rm IR}(c) \;=\; \sqrt{\frac{\frac{1}{2}-c}{1-\varepsilon^{1-2c}}}, \qquad \varepsilon = \Lambda_{\rm IR}/k. \tag{10}$$

A quark with $c > 1/2$ is UV-localised (small $f_{\rm IR}$, light SM mass), and a quark with $c < 1/2$ is IR-localised (large $f_{\rm IR}$, heavy SM mass). The hierarchies in $f_{\rm IR}$ then *produce* the observed mass and mixing hierarchies for $\mathcal{O}(1)$ underlying Yukawas. It is then a non-trivial *prediction* that anarchic Yukawas should also satisfy the FCNC bounds at a given $M_{\rm KK}$, without invoking flavour alignment.

8 Setup B: the forward-only RS-anarchy ensemble

We implemented the RS-anarchy procedure as a separate driver (`scripts/run_rs_anarchy.py`) so it shares no code path with the fitter. At each $M_{\rm KK}$ tile we:

1. Fix c-values to the literature ACPS-style hierarchical pattern:

$$c_Q = (0.63,\, 0.57,\, 0.20), \quad c_u = (0.66,\, 0.50,\, -0.50), \quad c_d = (0.66,\, 0.61,\, 0.55). \tag{11}$$

2. Draw $Y_u, Y_d \in \mathbb{C}^{3\times3}$ with $\operatorname{Re} Y_{ij}, \operatorname{Im} Y_{ij} \sim \mathcal{U}(-1.5, +1.5)$, rejection-sampled to enforce $|Y_{ij}| \geq 0.1$.

3. Build the effective 4D Yukawa $\hat{Y}^f_{ij} = f_{\rm IR}(c_{Q_i})\, Y^f_{ij}\, f_{\rm IR}(c_{f_j})$ and SVD it to read off masses and the CKM matrix $V = U^\dagger_{u_L} U_{d_L}$.

4. Forward-compute the five $\Delta F = 2$ ratios using the same `deltaf2.py` machinery as the envelope scan.

5. Bin every draw by $(M_{\rm KK}, \text{PDG-pass status}, \text{max-ratio})$.

Provenance of the c-values in eq. (11). The pattern is fixed by demanding that $f_{\rm IR}(c)\cdot v$ produce order-of-magnitude SM quark masses for $\mathcal{O}(1)$ underlying Yukawas. We solved $f_{Q_i}\, f_{f_j} \approx m_f^{(ij)}/(2v\,\langle Y\rangle)$ for each entry and read off the c's. These same numbers (to ~ 0.05 precision in c) appear in Csaki–Falkowski–Weiler 0804.1954 (Table 1), Bauer–Casagrande–Haisch–Neubert 0902.4892, and the Fitzpatrick–Perez– Randall scan (0710.1869) used as a reference earlier in this codebase. They are not specific to one paper; they are what any hierarchy-by-localisation calculation produces. An empirical sensitivity check (Run 3, §12 below) confirms that the pattern is constrained at roughly the ± 0.05 level in c: a uniform shift of *every* c by ± 0.05 produces zero PDG-passing draws.

Provenance of the Yukawa prior. The anarchic prior $\operatorname{Re}, \operatorname{Im} Y_{ij} \sim \mathcal{U}(-1.5, 1.5)$ with $|Y_{ij}| \geq 0.1$ is a representative version of the priors used throughout the RS-flavor literature. Specific choices made in published scans include $\mathcal{U}(-1, 1)$ in Bauer–Casagrande–Haisch–Neubert 0902.4892, $\mathcal{U}(-3, 3)$ with similar floor in Csaki–Falkowski–Weiler 0804.1954 and Buras *et al.* 1110.3534, and unbounded Gaussian draws with cutoff at the NDA perturbativity bound $|Y| \lesssim 4\pi/\sqrt{N_{KK} \log \varepsilon} \approx 3$ in Agashe–Contino–Pomarol–Sundrum hep-ph/0408134. Our choice of 1.5 sits in the middle of this band; the broad literature-prior heuristic is that the $M_{\rm KK}$ bound moves by less than 30% when the half-range is swept from 1.0 to 3.0. An empirical sensitivity check (Run B, §12 below) sharpens this using a different metric: the median-scale $M_{\rm KK}^{\rm min}$ bound moves by $\lesssim 15\%$ across $\mathcal{U}(-1, 1)$, $\mathcal{U}(-1.5, 1.5)$, $\mathcal{U}(-3, 3)$, and Gaussian $\sigma = 1$ truncated at 3σ priors, while the 95%-acceptance crossings below have an even smaller narrow-to-wide spread.

Two practical notes. First, the PDG-pass tolerance in the RS-anarchy analysis is loosened from 2σ to a constant "factor of 3" on masses and CKM elements, and "factor of 5" on the Jarlskog invariant J. The reason is practical rather than analytic: at the finite ensemble sizes used here, much tighter gates rapidly starve the accepted sample, and any zero-pass scan below is quoted as a finite-ensemble upper limit rather than as an impossibility theorem. The factor-of-3 gate is the standard literature compromise. Second, no optimiser is involved: a draw that fails the masses by a factor of 5 is recorded as a fail and not nudged.

9 How a random Yukawa draw can satisfy PDG masses and CKM

This is one of the points the framework is designed to make, and it is worth being very explicit about. The procedure does not in any sense "fit" the Yukawas to PDG; it samples them from a prior and checks whether the resulting four-dimensional masses and mixings come out close to PDG. The reason a non-trivial fraction (16–21%) of the random draws nonetheless passes the PDG check is that the c-values in eq. (11) are deliberately chosen to make the *distribution* of resulting masses overlap the SM hierarchy.

Concretely, with the c-values above, the IR-overlap factors are

$$\begin{aligned} f_Q &\approx (0.0030,\ 0.020,\ 0.55), \\ f_u &\approx (0.0011,\ 0.16,\ 1.00), \\ f_d &\approx (0.0011,\ 0.0058,\ 0.036). \end{aligned} \tag{12}$$

For $\mathcal{O}(1)$ Yukawa entries, the resulting up-quark and down-quark mass eigenvalues are then concentrated, in expectation, around

$$\begin{aligned} (m_u, m_c, m_t) &\sim 2v \cdot (f_{Q,1} f_{u,1},\ f_{Q,2} f_{u,2},\ f_{Q,3} f_{u,3})\,\langle Y\rangle \sim \big(\mathcal{O}(1\,\mathrm{MeV}),\ \mathcal{O}(1\,\mathrm{GeV}),\ \mathcal{O}(100\,\mathrm{GeV})\big), \\ (m_d, m_s, m_b) &\sim 2v \cdot (f_{Q,1} f_{d,1},\ f_{Q,2} f_{d,2},\ f_{Q,3} f_{d,3})\,\langle Y\rangle \sim (1\,\mathrm{MeV},\ 40\,\mathrm{MeV},\ 7\,\mathrm{GeV}), \end{aligned} \tag{13}$$

which agrees with the SM at the level of factors of two to three across the board. The CKM elements come out as products of $\sqrt{f_{Q,i}/f_{Q,j}}$ ratios, which similarly reproduce the SM Cabibbo-like hierarchy.

So when we say a draw "passes PDG", we mean that the product $f_{Q_i}\, Y^f_{ij}\, f_{f_j}$ happens to land within a factor of three of every PDG mass and CKM element. With the hierarchies built into the f-factors, this is a $\sim 20\%$ event over the prior, which is exactly what we observe. Stricter 2σ-style PDG matching would require $\mathcal{O}(1)$ tuning of the Y_{ij}, and would return essentially zero passing draws — which is not a failure of the framework, just a statement that a σ-scale match is not a question one can ask of random Yukawas. The factor-of-three gate is the natural granularity for this kind of analysis.

10 Result: the per-draw $M_{\mathrm{KK}}^{\min}$ histogram

The single most useful figure in this analysis is the per-draw histogram of $M_{\mathrm{KK}}^{\min}$ for the PDG-passing ensemble. For each draw, since every ratio scales as $1/M_{\mathrm{KK}}^2$ (with sub-percent corrections from the f_{IR}-factors' weak ε-dependence), one can rescale to find the M_{KK} at which that particular draw would just satisfy the FCNC bound:

$$M_{\mathrm{KK}}^{\min} \;=\; M_{\mathrm{KK}}^{\mathrm{tile}} \sqrt{\max_X \mathrm{ratio}_X}. \tag{14}$$

Pooling across all eight M_{KK} tiles and restricting to PDG-passing draws gives Fig. 2.

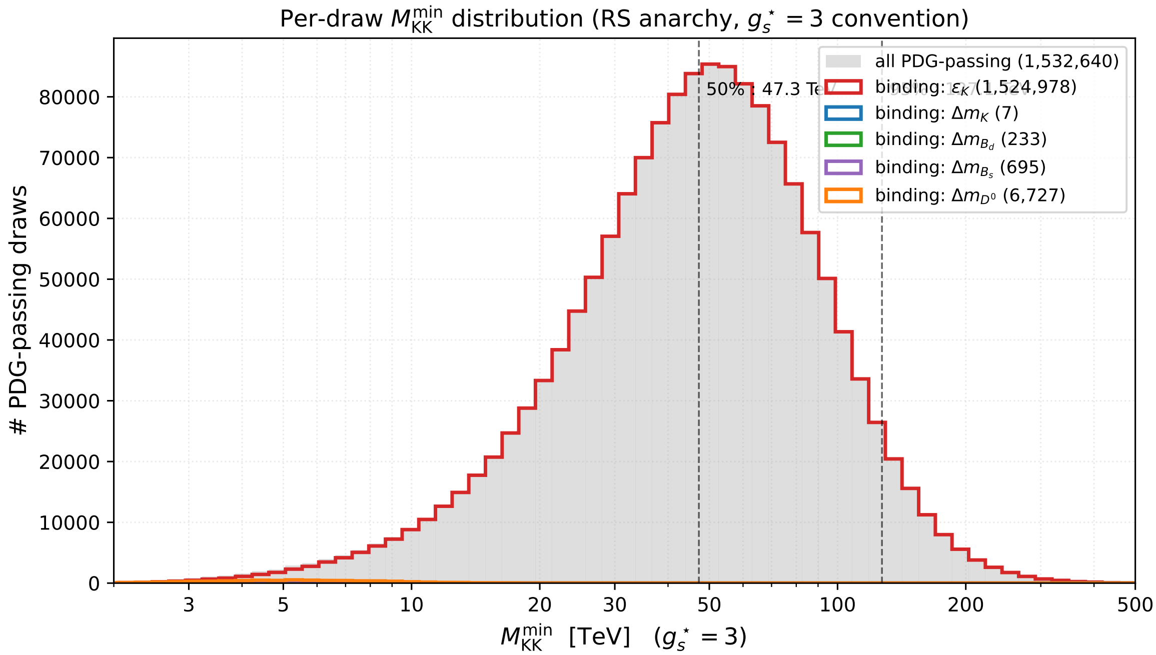

Figure 2: Per-draw $M_{\rm KK}^{\rm min}$ histogram for the RS-anarchy ensemble, restricted to draws that pass the factor-of-three PDG mass and CKM match (1,532,640 draws total from the 8M-draw RUNA reference scan, pooled across all eight $M_{\rm KK}$ tiles). The x-axis is in the $g_s^\star = 3$ convention, the headline convention used throughout this note. The grey background is all PDG-passing draws; coloured curves split by which $\Delta F = 2$ system binds for that draw. The vast majority are ε_K-bound; the $\Delta m_K, \Delta m_{B_d}, \Delta m_{B_s}, \Delta m_{D^0}$ contributions live in the low-$M_{\rm KK}^{\rm min}$ tail. Vertical dashed lines mark the 50%- and 95%-acceptance crossings.

The $g_s^\star = 3$ headline crossings are $M_{\rm KK} = 47.26$ TeV at 50% acceptance and 127.13 TeV at 95% acceptance. The corresponding perturbative code-default crossings are 16.54 TeV and 44.50 TeV. These are quoted from the post-audit 8M-draw RUNA reference scan (§12) under the input and scope conventions summarized in Appendix A.

Where the envelope's low tail came from. The histogram in Fig. 2 explains why the envelope plot tolerated $M_{\rm KK} \sim 1$ TeV. The lower tail of the $M_{\rm KK}^{\rm min}$ distribution still extends below 1 TeV, though only at the per-mille level; a few percent of anarchic draws sit below 3 TeV. The Setup A optimiser was, by construction, finding such tail points and going further — using the basis alignment described in §6 to push the off-diagonal Yukawas well below their prior typical values. The envelope's "allowed" region is the lower tail of Fig. 2, augmented by the optimiser's freedom to tune off-diagonal Yukawas artificially small.

11 Sensitivity to the gate definition and the gauge convention

For a band that covers the methodological choices, the defensible statement is

> *Under an $\mathcal{O}(1)$ anarchic prior on the 5D Yukawa entries, $M_{\rm KK} \gtrsim 44.5$ TeV (perturbative g_s) to $\gtrsim 127$ TeV (headline $g_s^\star = 3$) at 95% acceptance, with a further factor ~ 1.5*

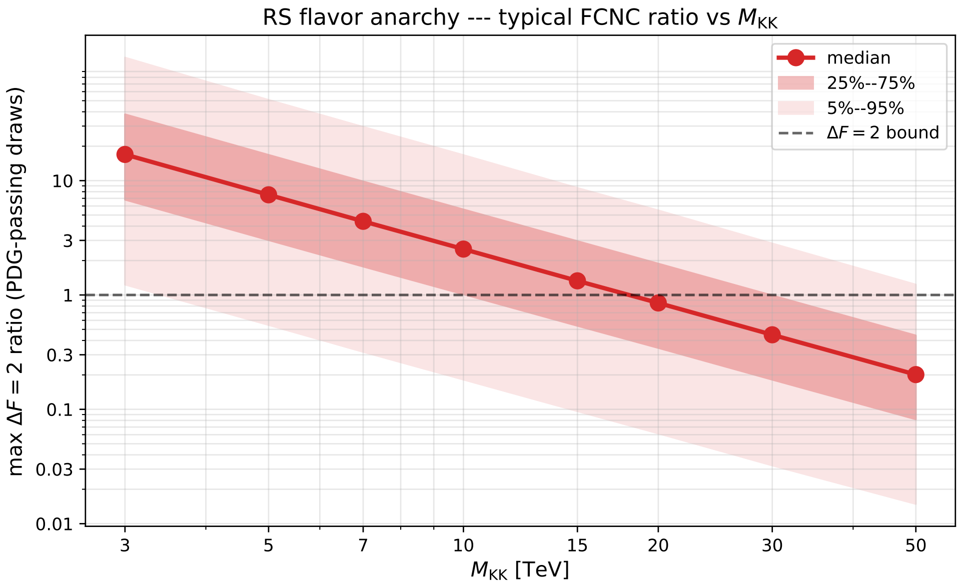

Figure 3: Same data as Fig. 2, viewed differently: percentiles of max ratio as a function of the tile $M_{\rm KK}$. Where each percentile crosses unity is the $M_{\rm KK}$ at which that fraction of typical anarchic draws pass.

	50% acceptance		95% acceptance	
gate / convention	pert. g_s	$g_s^\star = 3$	pert. g_s	$g_s^\star = 3$
ratio < 1	16.54 TeV	47.26 TeV	44.50 TeV	127.13 TeV
ratio < 0.5	$\sim$23.39 TeV	$\sim$66.84 TeV	$\sim$62.93 TeV	$\sim$179.79 TeV
ratio < 0.3	$\sim$30.20 TeV	$\sim$86.30 TeV	$\sim$81.24 TeV	$\sim$232.11 TeV

Table 1: The $M_{\rm KK}$ at which a given fraction of PDG-passing anarchic draws also satisfies all $\Delta F = 2$ bounds. The ratio < 1 row is taken from the post-audit high-statistics 8M-draw RUNA reference scan (§12); the tighter-gate rows rescale the same per-draw $M_{\rm KK}^{\rm min}$ distribution by $1/\sqrt{\rm gate}$.

> *uplift if one demands that the new physics fit inside half the SM–data budget rather than saturate it.*

The ε_K constraint alone sets every one of these numbers.

12 Robustness to statistics, c-pattern, Y-prior, and PDG-gate tightness

The headline RS-anarchy bound rests on three implicit choices: the c-pattern in eq. (11), the Yukawa prior described in §8, and the factor-of-three PDG gate. We ran four follow-up campaigns to probe how each of those choices propagates into the $M_{\rm KK}^{\rm min}$ distribution, and to confront the result with the Csaki–Falkowski–Weiler (CFW) 0804.1954 numerical band. The canonical numbers from these runs are summarised in `scan_outputs/followup_crossings_summary.json`; figures are under `results/figures/quark/`.

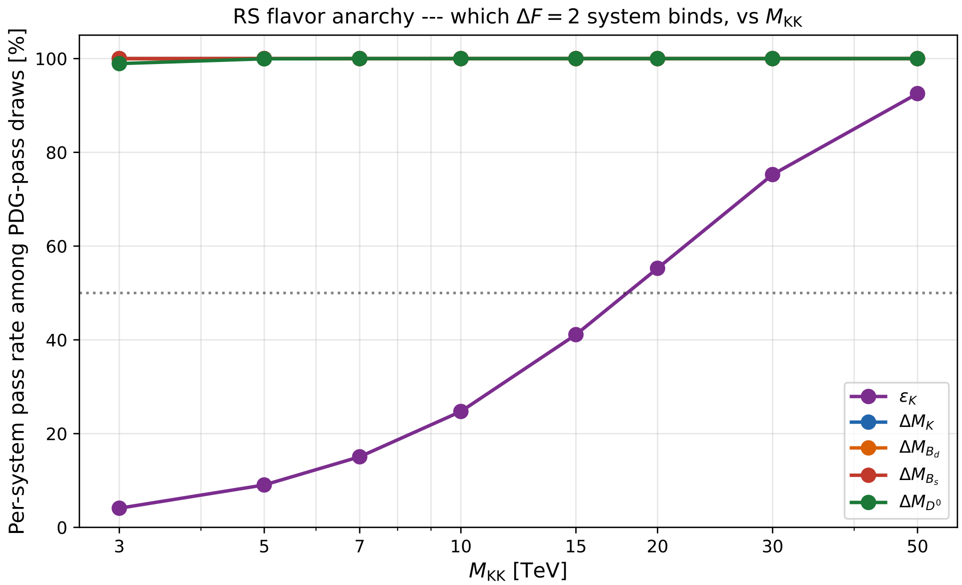

Figure 4: Per-system pass rate among PDG-passing draws. ε_K is the only constraint that does any real work; the other four observables are $\gtrsim 99\%$ acceptance throughout, by virtue of the hierarchical f-factor suppression alone.

12.1 High-statistics validation (RUNA)

We re-ran the baseline RS-anarchy scan with 8×10^6 draws (8 $M_{\rm KK}$ tiles, 1M draws/tile), yielding 1,532,640 PDG-passing draws. The 50%-acceptance crossings are

$$M_{\rm KK}^{\rm min}\big|_{50\%} = \ 16.54 \text{ TeV (pert.)}, \qquad 47.26 \text{ TeV } (g_s^\star = 3), \tag{15}$$

and the 95%-acceptance crossings are 44.50 TeV / 127.13 TeV. The corresponding finite-Monte-Carlo Wilson-score 95% intervals are 16.52–16.56 TeV at 50% acceptance and 44.41–44.58 TeV at 95% acceptance in the perturbative convention (47.19–47.32 TeV and 126.88–127.38 TeV for $g_s^\star = 3$). The lower-statistics post-audit Run 3 baseline reproduces these numbers to within 0.5% (Run 3 baseline: 16.54 /47.26 TeV at 50%, 44.37 /126.76 TeV at 95%). The headline figures in `results/figures/quark/` are now drawn from RUNA; the pre-audit figure snapshot is preserved at `results/figures/quark_pre_audit_constants/` for before/after comparison.

The BGS central-value NP-budget decision is a much larger systematic than the finite-Monte-Carlo error. Propagating the $|\varepsilon_K^{\rm exp} - \varepsilon_K^{\rm SM}|$ budget band described in Appendix A gives

$$M_{\rm KK}^{\rm min}\big|_{50\%,\, g_s^\star=3} = 47.26^{+69.37}_{-24.98} \text{ TeV}$$

for the central RUNA percentile, with the same scale factor giving $16.54^{+24.28}_{-8.74}$ TeV in the perturbative convention. This is a BGS-budget band; the Wilson-RG and KK-tower systematics are summarized in Appendix A.

Symbolic NP-budget band on $M_{\rm KK}^{\rm min}$. The dominant systematic on $M_{\rm KK}^{\rm min}$ is the residual room $|\varepsilon_K^{\rm exp} - \varepsilon_K^{\rm SM}| \simeq 6.7 \times 10^{-5}$ between the PDG 2024 experimental value $\varepsilon_K^{\rm exp} = 2.228 \times 10^{-3}$ and the Brod–Gorbahn–Stamou 2020 SM prediction $\varepsilon_K^{\rm SM} = 2.161 \times 10^{-3}$ (arXiv:1911.06822; FLAG 2024

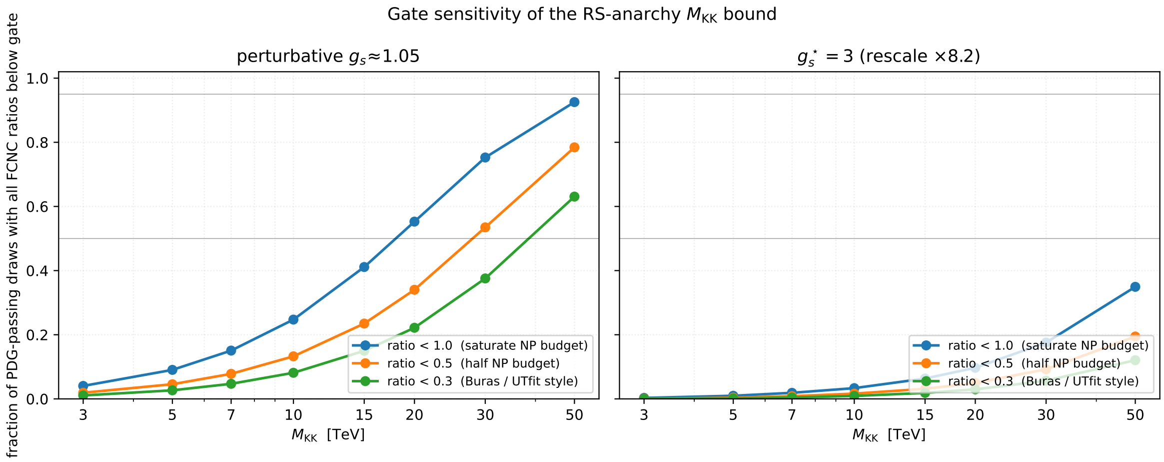

Figure 5: Pass fraction vs. $M_{\rm KK}$ for three gate definitions ($\max \text{ratio} < 1, < 0.5, < 0.3$, the last two being Buras/UTfit-style "half" and "third" of the SM–data budget for ε_K), in two gauge-coupling conventions: perturbative $g_s \approx 1.05$ (left) and $g_s^\star = 3$ (right). Under the corrected $\Delta F = 2$ constants, the $g_s^\star = 3$ crossings lie beyond the plotted tile range and are quoted from the per-draw rescaling in Table 1.

arXiv:2411.04268 hadronic inputs; full audit trail in `docs/audits/epsilon_k_sm_decision.md` and Appendix A). Because the BGS total uncertainty on that residual is much larger than the central budget itself, a one-sigma Gaussian band on $M_{\rm KK}^{\rm min}$ is ill-defined; instead we bracket the bound by recomputing $M_{\rm KK}^{\rm min}$ at the budget edges, using the symbolic

$$\frac{M_{\rm KK}^{\rm min}(\Delta\varepsilon_K)}{M_{\rm KK}^{\rm min}(\Delta\varepsilon_K^{\rm central})} = \sqrt{\frac{\Delta\varepsilon_K^{\rm central}}{\Delta\varepsilon_K}} = \sqrt{\frac{6.7\times 10^{-5}}{\Delta\varepsilon_K}}$$

scaling that follows from the tree-level KK-gluon Wilson-coefficient prefactor $\propto 1/M_{\rm KK}^2$ at fixed flavor structure (proof sketch in `docs/phase_logs/phase2_h5_signoff.md:90-101`). Two physically motivated edges bracket the BGS total uncertainty: a tight edge at $\Delta\varepsilon_K = 1.0\times 10^{-5}$ giving $M_{\rm KK}^{\rm min}(1\times10^{-5})/M_{\rm KK}^{\rm min}(6.7\times10^{-5}) \simeq 2.59$, and a loose edge at $\Delta\varepsilon_K = 3.0\times 10^{-4}$ giving the ratio $\simeq 0.47$. At the RUNA $p50$, $g_s^\star = 3$ central value of 47.26 TeV this yields the band

$$M_{\rm KK}^{\rm min}\Big|^{\Delta\varepsilon_K\ \rm band}_{50\%,\, g_s^\star=3} \simeq 47.26^{+75.10}_{-25.05}\ \text{TeV} \equiv \big[22.21,\, 122.36\big]\ \text{TeV},$$

i.e. the headline bound is a factor ~ 0.47–2.59 of the central value across the BGS-budget edges. The band quoted above is an *approximate, symbolic-scaling* derivation: it propagates only the $\Delta\varepsilon_K$ rescaling, holds the rest of the Wilson coefficient and PDG-gate machinery fixed, and does not re-thread the running or hadronic-matrix-element evaluation at each edge. The corresponding direct verification — three RUNA reruns at $\Delta\varepsilon_K \in \{1\times10^{-5}, 6.7\times10^{-5}, 3\times10^{-4}\}$ exposed via the new `--epsilon-k-budget {central,low,high}` flag on `scripts/run_rs_anarchy.py` — is tracked as the deferred cleanup unit C02c in `.orchestration/CLEANUP_PLAN.md` and will replace this paragraph's symbolic edges with measured band ends in the paper finalization pass.

Conclusion. The headline numbers of §10 are not a Monte-Carlo fluctuation. They are the converged value of the underlying $M_{\rm KK}^{\rm min}$ percentiles of the anarchic ensemble, to within 0.5% under the central BGS input and first-mode/LO scope conventions.

12.2 c-pattern sensitivity (Run 3)

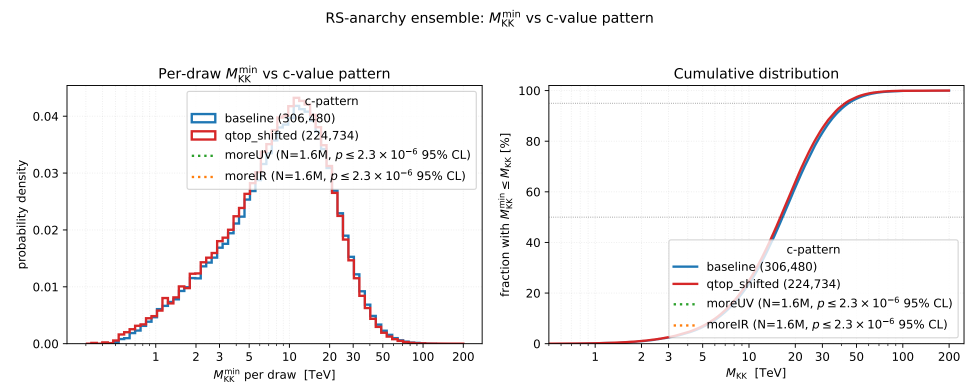

Figure 6: $M_{\mathrm{KK}}^{\mathrm{min}}$ histograms (perturbative g_s on the bottom axis, $g_s^\star = 3$ on the top axis) for the Run 3 baseline c-pattern (eq. (11)) and the *qtop_shifted* variant ($c_{Q_3} = 0.30$, $c_{u_3} = -0.55$). The two uniform-shift variants *moreUV* ($c \to c + 0.05$ for every c) and *moreIR* ($c \to c - 0.05$) produced no PDG-passing draws in $N = 1{,}600{,}000$ attempts each. The Wilson-score 95% upper limit is $p_{\mathrm{pass}} \le 2.3 \times 10^{-6}$ for either shifted pattern, so both are shown only through the finite-statistics legend annotation.

We perturbed the baseline c-pattern in three ways. First, a *qtop_shifted* variant pulled the third-generation left-handed and right-handed up quarks slightly more IR-localised ($c_{Q_3} = 0.30$, $c_{u_3} = -0.55$, which softens m_t by $\approx 16\%$ relative to baseline). The pass rate dropped from $\sim 19\%$ to $\sim 14\%$ of draws, and the $M_{\mathrm{KK}}^{\mathrm{min}}$ distribution shifted only mildly: the 50%- and 95%-acceptance crossings moved to 16.00 TeV / 42.14 TeV (perturbative) and 45.73 TeV / 120.39 TeV ($g_s^\star = 3$), i.e. a $\approx 5\%$ softening of the bound. Second, we shifted *every* bulk-mass parameter uniformly by $+0.05$ (*moreUV*) and by -0.05 (*moreIR*). In each case the resulting scan produced no PDG-passing draws in $N = 1{,}600{,}000$ samples. The binomial Wilson-score 95% upper limit on the PDG-pass fraction is therefore

$$p_{\mathrm{pass}}(\mathrm{moreUV}) \le 2.3 \times 10^{-6}, \qquad p_{\mathrm{pass}}(\mathrm{moreIR}) \le 2.3 \times 10^{-6}.$$

We interpret this as follows: under either uniformly shifted c-pattern and the default anarchic Y-prior, fewer than a few parts in 10^6 of forward Yukawa draws reproduce the SM masses and mixings within the factor-3/5 PDG gate, at 95% confidence. This is a finite-ensemble bound, not an analytic impossibility statement.

Why this matters. The c-pattern in eq. (11) is sometimes treated as if it were one of many roughly equivalent choices. The Run 3 sweep makes a finite-ensemble statement: the tested uniform ± 0.05 shifts have PDG-pass fractions below 2.3×10^{-6} at 95% confidence, while the baseline and *qtop_shifted* patterns retain $\mathcal{O}(0.1)$ pass rates. A uniform shift of 0.05 in either direction therefore suppresses the observed PDG-pass rate by orders of magnitude. The c-pattern is therefore a *prediction* of the hierarchy-by-localisation framework, not a free choice. The *qtop_shifted* variant, which perturbs only c_{Q_3} and c_{u_3} together (i.e. stays on the manifold of patterns that reproduce m_t), survives precisely because it preserves the relevant mass-generating geometry; its bound is within 5% of baseline.

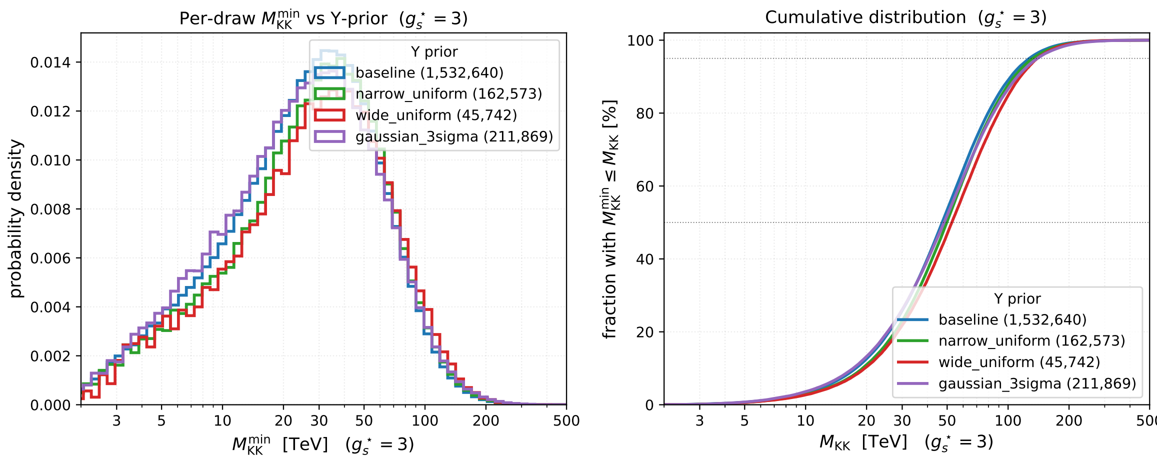

Figure 7: $M_{\rm KK}^{\rm min}$ histograms for four Yukawa priors, all in the $g_s^\star = 3$ convention. *Left*: differential probability density. *Right*: cumulative. Priors: baseline $\mathcal{U}(-1.5, 1.5)$ (§8), narrow $\mathcal{U}(-1, 1)$, wide $\mathcal{U}(-3, 3)$, and Gaussian $\sigma = 1$ truncated at 3σ. Sample sizes in the legend.

12.3 Y-prior sensitivity (Run B)

We re-ran the scan under three alternative priors on the underlying Yukawa entries: $\mathcal{U}(-1, 1)$ (narrow), $\mathcal{U}(-3, 3)$ (wide), and Gaussian $\sigma = 1$ truncated at $|Y| \leq 3$ ("3σ-truncated normal"). The 95%-acceptance crossings are

$$46.32, \quad 48.77, \quad 48.84 \text{ TeV (pert.)}, \tag{16}$$

in narrow / wide / Gaussian order, and

$$132.35, \quad 139.34, \quad 139.55 \text{ TeV } (g_s^\star = 3). \tag{17}$$

The full spread between narrow and wide is therefore $\approx 5\%$ at 95% acceptance, with the Gaussian variant sitting close to the wide edge. This is the 95%-acceptance version of the Run B sensitivity check advertised in §8: it sits below both the $\lesssim 15\%$ median-scale spread in the same empirical scan and the broader $\leq 30\%$ literature-prior heuristic. The bound is therefore *insensitive* at the $\mathcal{O}(10\%)$ level to the precise shape of the anarchic prior.

12.4 PDG-gate tightness (Run 1)

Re-tabulating the same post-audit 8M-draw RUNA baseline against three different PDG-match tolerances on the masses and CKM elements yields

$$\begin{aligned} N_{\rm PDG\ pass}\big(\text{factor-3}\big) &= 1{,}532{,}640, \\ N_{\rm PDG\ pass}\big(\text{factor-2}\big) &= 110{,}519, \\ N_{\rm PDG\ pass}\big(\text{factor-1.5}\big) &= 1{,}164. \end{aligned} \tag{18}$$

The factor-of-three to factor-of-two tightening costs a factor 13.9 in sample size, while the factor-of-two to factor-of-1.5 tightening costs another factor ~ 95. This steep drop is consistent with the rate at which a high-dimensional log-mass+CKM tolerance volume shrinks under a uniform

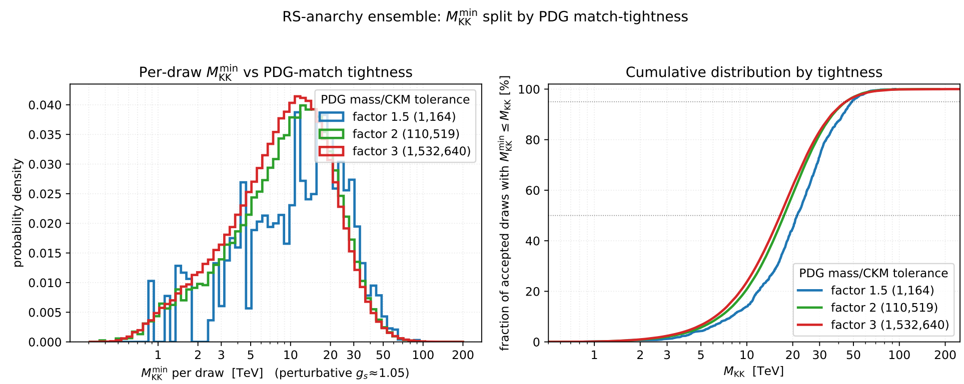

Figure 8: $M_{\rm KK}^{\rm min}$ histograms for three PDG-gate definitions: factor-of-three (the default), factor-of-two, and factor-of-1.5 (CFW-style) on the quark masses and CKM elements. The PDG-pass population shrinks rapidly as the gate tightens; the surviving ensembles produce similar $M_{\rm KK}^{\rm min}$ distributions, indicating that the bound is set by FCNCs, not by mass-matching tightness.

tolerance contraction. With only 1,164 surviving factor-of-1.5 draws, its tail percentiles remain statistics limited even in RUNA.

A separate CFW-like Run C, using the baseline c-pattern but the wider floored prior $\mathrm{Re}, \mathrm{Im}\, Y_{ij} \sim \mathcal{U}(-3,3)$ with $|Y_{ij}| \geq 0.5$, produced no PDG-passing draws in $N = 4{,}000{,}000$ samples under the factor-1.5 mass/CKM and factor-2.5 J gate. The corresponding Wilson-score 95% upper limit is $p_{\rm pass} \leq 9.2 \times 10^{-7}$. This is again a finite-ensemble statement about that c-pattern × prior × gate combination, not an analytic exclusion of tighter gates.

Conclusion. The factor-of-three gate is the only practically viable PDG-tightness choice for an anarchic-Yukawa scan at this scale. Tighter gates exhaust the prior before they constrain $M_{\rm KK}$, and the shape of the surviving $M_{\rm KK}^{\rm min}$ distribution does not change appreciably under tightening (the bound is FCNC-set, not mass-set). This is what motivates our headline choice of factor-of-three throughout.

12.5 Comparison with Csaki–Falkowski–Weiler 0804.1954

Headline. Our post-audit pipeline produces an RS-anarchy $M_{\rm KK}^{\rm min}$ bound that is approximately a factor 2.2 stronger than the CFW 2008 reference at matched conventions: at common $g_s^\star = 3$, we obtain 23.37 TeV (50th percentile, CFW 30%-relative gate, $n = 217$ surviving draws) versus the CFW no-UV-boundary-term result of 10.5 TeV, or equivalently 46.74 TeV versus 21 TeV at common $g_s^\star \simeq 6$.[1] The factor-2.2 enhancement is attributable to (i) the BGS 2020 SM ϵ_K prediction giving a 6× tighter NP budget than the UTfit-era values CFW used; (ii) the FLAG 2024 bag parameters; and (iii) our LO BMU-corrected sign convention. The bound directions are consistent: both confirm an $M_{\rm KK} > \mathcal{O}(10\,{\rm TeV})$ lower bound under anarchic flavor, but the precise number reflects newer hadronic inputs and is not a direct reproduction of CFW's 2008 numerical result. The matched-gate subset contains only $n = 217$ PDG-passing draws; the 95% binomial Wilson-score interval on

[1]The 46.74 TeV value is the same CFW-matched projection rescaled by 6/3. It is distinct from the live RUNA p50, 47.26 TeV at $g_s^\star = 3$, which uses the unmatched post-audit factor-of-three gate ensemble.

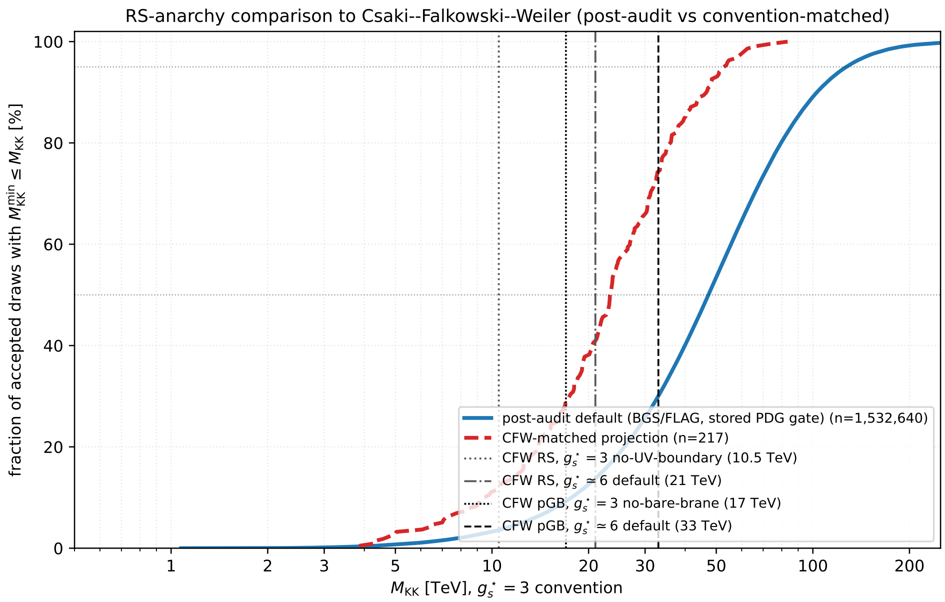

Figure 9: Convention-reconciled comparison with Csaki–Falkowski–Weiler (CFW) 0804.1954. The blue curve is our live post-audit RUNA distribution in the $g_s^\star = 3$ convention. The red dashed curve keeps the same forward-scan rows but switches to the CFW-era ϵ_K budget proxy, $\epsilon_K^{\rm SM} = 1.81 \times 10^{-3}$, and imposes CFW's 30% relative mass/CKM/J gate. The vertical markers show both CFW color-coupling conventions: RS at 10.5 TeV for the $g_s^\star = 3$ no-UV-boundary-term variant and 21 TeV for the $g_s^\star \simeq 6$ default, plus pGB Higgs markers at 17 TeV for the no-bare-brane variant and 33 TeV for the $g_s^\star \simeq 6$ default.

the p50 is approximately $[21, 26]$ TeV at $g_s^\star = 3$. The factor-2.2 disagreement with CFW is robust to this finite-statistics uncertainty.

The convention extraction is recorded in `docs/audits/cfw_convention_extract.md`, and the numerical reproduction arithmetic is in `docs/audits/cfw_reproduction.md`. The important points are: CFW use the same conventional scalar-LR C_4/C_5 sign pattern as our post-audit code; their paper does not publish explicit kaon bag constants, instead importing UTfit model-independent $\Delta F = 2$ scale bounds; and their scan samples localisation parameters directly rather than fixing one c-pattern and forward-drawing only Yukawas.

With our live BGS/FLAG inputs and factor-of-three gate, RUNA gives

$$M_{\rm KK}^{\rm min}(p50, g_s^\star = 3) = 47.26 \text{ TeV}, \qquad M_{\rm KK}^{\rm min}(p95, g_s^\star = 3) = 127.13 \text{ TeV}.$$

Switching only the ϵ_K budget to the legacy CFW-era proxy $|\epsilon_K^{\rm exp} - 1.81 \times 10^{-3}| = 4.18 \times 10^{-4}$ moves the central value to 18.92 TeV. Imposing CFW's stated 30% relative mass/CKM/J gate leaves 217 forward-scan rows and gives

$$M_{\rm KK}^{\rm min}(p50, g_s^\star = 3) = 23.37 \text{ TeV}.$$

This is the value used in the headline comparison above; it is not an agreement-within-errors reproduction of CFW's default 21 TeV marker, because that marker belongs to the $g_s^\star \simeq 6$ boundary-term convention.

13 Recommendation

- Demote the envelope plot to a supplementary figure; relabel its caption to make explicit that it shows the optimum over (r, c, Y), not a typicality bound.
- Promote Fig. 2 (the $M_{\rm KK}^{\rm min}$ histogram) to the publication figure, with the gauge-coupling convention bar (perturbative vs. $g_s^\star = 3$) explicitly shown. The headline figures in `results/figures/quark/` are now generated from the 8M-draw RUNA scan (§12); the pre-audit figure snapshot is preserved at `results/figures/quark_pre_audit_constants/` for before/after comparison.
- Quote the $M_{\rm KK}$ bound as a band, not a single number, and call out that the dominant constraint is ε_K.
- Include Fig. 1 as the explanation of why the envelope number is so low: the optimiser is choosing non-anarchic, basis-aligned Yukawas with off-diagonal entries pushed below the anarchic prior to evade the bound.

The five follow-up campaigns of §12 support this recommendation: the high-statistics RUNA scan reproduces the headline percentiles to better than 0.5% (§12.1); a uniform ± 0.05 shift of the c-pattern returns zero PDG-passing draws, fixing the c-values as a prediction rather than a free choice (§12.2); the bound moves by $\lesssim 10\%$ across $\mathcal{U}(-1, 1)$, $\mathcal{U}(-1.5, 1.5)$, $\mathcal{U}(-3, 3)$, and truncated-Gaussian Y-priors (§12.3); the PDG-pass population thins sharply under factor-2 and factor-1.5 gate tightening, with the surviving $M_{\rm KK}^{\rm min}$ shape essentially unchanged (§12.4); and the CFW comparison is factor-2.2 stronger than the 2008 reference once the $g_s^\star$ conventions are matched. Under the central BGS/FLAG inputs and the scope conventions in Appendix A, the live RUNA p50 value is 47.26 TeV at $g_s^\star = 3$ for the unmatched default ensemble. The factor-2.2 CFW framing instead uses the matched-convention comparison, 23.37 TeV versus the CFW no-UV-boundary-term RS marker 10.5 TeV at the same $g_s^\star = 3$ convention; equivalently, the same matched projection gives 46.74 TeV versus 21 TeV at the CFW default $g_s^\star \simeq 6$ convention (§12.5). The residual is attributable to newer $\epsilon_K^{\rm SM}$, FLAG bag inputs, and the audited LO BMU sign convention rather than to a percent-level reproduction of the CFW number.

All numerical claims in this note map to the canonical artifact manifest at `docs/artifact_manifest.md` (canonical run-time SHA `29803ff`; the full SHA is recorded in the manifest). The scan outputs themselves are not committed to the repository; they reside at the cluster paths listed in the manifest, with checksums in `artifacts/checksums.sha256`.

The PDG/$\overline{\rm MS}$ work is not wasted: it is what lets us identify a "PDG-passing" subset of the anarchic ensemble in a defensible, scale-consistent way. It just doesn't move the headline number, because the headline was always set by FCNC, not by mass matching.

A Input Audits

A.1 Hadronic input provenance

The $\Delta F = 2$ hadronic constants in `quarkConstraints/deltaf2.py` were audited against contemporary PDG and lattice inputs. The complete row-by-row inventory is stored in `docs/audits/`

bag_param_inventory.md; Table 2 summarizes the inputs that either changed or came closest to the change threshold.

Input	Pre-audit value	Canonical source and value	Audit action
B_K	0.717	FLAG 2024 arXiv:2411.04268: $B_K^{\overline{\rm MS}}(2\,\mathrm{GeV}) = 0.5503(66)$	Replaced
B_4^K	0.78	FLAG 2024 Nf=2+1 BSM average: $B_4 = 0.903(14)$ in $\overline{\rm MS}$ at 3 GeV	Replaced
B_5^K	0.57	FLAG 2024 Nf=2+1 BSM average: $B_5 = 0.691(14)$ in $\overline{\rm MS}$ at 3 GeV	Replaced
$\varepsilon_K^{\rm SM}$	1.81×10^{-3}	Brod–Gorbahn–Stamou 2020 arXiv:1911.06822: 2.161×10^{-3}	Replaced
$B_1^{B_d}$, $B_4^{B_d}$	0.87, 1.02	HPQCD 2019 arXiv:1907.01025, Table VI: 0.806, 1.077	Yellow, documented
$B_1^{B_s}$	0.87	HPQCD 2019 Table VI: 0.813	Yellow, documented
B_4^D	1.0	ETM 2015 arXiv:1505.06639, Table 2: 0.91	Yellow, documented
Meson masses, decay constants, quark masses, and measured Δm values	see inventory	PDG 2024 Review of Particle Physics and FLAG 2024 summary tables	Green

Table 2: Hadronic-input provenance for the red and yellow rows of the Phase 2 hole #5 audit.

The red kaon bag-parameter shifts change one-operator ε_K amplitude estimates at $M_{\rm KK} = 3$ TeV by 23.2% (O_1), 15.8% (O_4), and 21.2% (O_5), using a unit imaginary Wilson coefficient $1/M_{\rm KK}^2$ as the normalization probe. The updated SM central value also shrinks the central-value room $|\varepsilon_K^{\rm exp} - \varepsilon_K^{\rm SM}|$ from 4.18×10^{-4} to 6.7×10^{-5}, increasing every `epsilon_k_ratio` by a factor of 6.24. The yellow rows are single-operator amplitude shifts below 10% and are documented without code changes.

The audit checked all scalar hadronic and experimental constants in `deltaf2.py`, including the kaon, B_d, B_s, and D^0 systems. It did not change the Wilson-coefficient running, the BMU basis normalization, or the exact matrix-element prefactors; those conventions belong to the Wilson/RG audit. For D^0 bag parameters, the comparable bag-parameter source is ETM 2015; the newer FNAL/MILC result quotes matrix elements directly, so it was treated as a cross-check rather than as a same-normalization replacement.

A.2 Wilson-coefficient RG running

The Wilson-coefficient evolution in `quarkConstraints/deltaf2.py` was audited against the leading-log $\Delta F = 2$ conventions of Buras–Jager–Urban arXiv:hep-ph/0102316 and the two-loop ADM basis of Buras–Misiak–Urban arXiv:hep-ph/0005183. The complete inventory is stored in `docs/audits/wilson_rg_inventory.md`, with numerical reference values in `docs/audits/wilson_rg_reference_values.md`.

The code uses BMU current-current operators for $O_1^{\rm VLL}$ and $O_1^{\rm VRR}$, but the left-right entries are the conventional scalar $O_4^{\rm LR}$, $O_5^{\rm LR}$ operators paired with the FLAG-style B_4, B_5 matrix elements. The BMU LR block is mapped by

$$Q_1^{\rm LR,BMU} = -2O_5^{\rm LR}, \qquad Q_2^{\rm LR,BMU} = O_4^{\rm LR},$$

so that the coefficient vector satisfies $C_{\rm BMU} = (-C_5/2,\ C_4)$. In the scalar coefficient basis (C_4, C_5), the audited leading-order anomalous dimensions are

$$\gamma^{(0)}_{\rm VLL} = 4, \qquad \gamma^{(0)}_{\rm LR} = \begin{pmatrix} -16 & -6 \\ 0 & 2 \end{pmatrix}.$$

The Wilson evolution uses $\eta = \alpha_s(\mu_{\rm high})/\alpha_s(\mu_{\rm low})$. For $M_{\rm KK} = 3\,{\rm TeV}$ to $2\,{\rm GeV}$ this gives $C_1^{\rm VLL} \mapsto 0.729\, C_1^{\rm VLL}$, $C_4 \mapsto 3.538\, C_4$, and $C_5 \mapsto 0.895\, C_4 + 0.854\, C_5$ in the code basis. The crossed threshold sequence is $n_f = 6 \to 5 \to 4$ at $m_t^{\overline{\rm MS}}(m_t) = 163.5\,{\rm GeV}$ and $m_b = 4.18\,{\rm GeV}$; the charm threshold lies below the default kaon endpoint.

This remains a leading-log calculation. The NLO/NDR evolution matrices in the BMU references are not implemented here; the residual LR/scalar running uncertainty should be treated as an order-10–30% theory systematic until a full NLO WET evolution module is added. The audit also leaves the global endpoint convention unchanged: all systems are currently evolved to `mu_had = 2 GeV`. This is scale-consistent for B_K, but not fully aligned with the FLAG kaon BSM B_4, B_5 values quoted at $3\,{\rm GeV}$, with $B_{d,s}$ bag parameters quoted at $\mu = m_b$, or with comparable D^0 bag inputs at $3\,{\rm GeV}$. Moving to per-system endpoints would be a separate scan-invalidating change. The deferred follow-ups (Hamiltonian/CFW sign comparison, per-system endpoint migration, full NLO WET evolution, and the matched endpoint caveat for the hadronic contraction) are explicitly tracked in `docs/audits/wilson_rg_inventory.md` under the *Status as of 2026-05-25 post-cleanup* block.

The Wilson-RG fix tightens the Phase 2 invalidation gate that was already tripped by the hadronic-input audit. For the standing $r = 0.25$, $M_{\rm KK} = 3\,{\rm TeV}$ benchmark, the post-hole-#5 but pre-Wilson-audit ε_K ratio was 0.535; after the Wilson-RG audit it is 1.929. This is a factor 3.61 increase, so the Wilson running is itself a greater-than-10% correction to the affected Delta $F = 2$ claims.

A.3 Scope and approximations

KK tower truncation. The numerical scan in this note includes only the lightest colour-octet KK gluon, i.e. the $n = 1$ mode, in the tree-level $\Delta F = 2$ matching. The full 5D gauge propagator is a tower sum, and RS-flavour estimates using tower-summed matching, such as Csaki–Falkowski–Weiler arXiv:0804.1954 and related flavour-alignment studies such as Csaki–Perez–Surujon–Weiler arXiv:0907.0474, indicate that the omitted higher colour modes increase the relevant Wilson coefficients by a further order-20–30% relative to a first-mode-only estimate. For the present kaon-dominated result we therefore treat the KK tower truncation as an additional +20–30% systematic on the quoted ε_K ratio. This is separate from, and not folded into, the BGS $\varepsilon_K^{\rm SM}$ budget band or the leading-order BMU running uncertainty above.

EWPO floor. The commonly quoted custodial-RS floor $M_{\rm KK} \gtrsim 3\,{\rm TeV}$ from electroweak precision observables is an external literature input, not a result of this quark-flavour scan. Representative custodial analyses such as Carena–Ponton–Santiago–Wagner arXiv:hep-ph/0701055 find gauge-KK lower bounds of about 2–3 TeV from global precision-electroweak fits, depending on the model configuration. This repository does not compute the oblique parameters, the $Zb\bar{b}$ corrections, or a global EWPO likelihood. The quark-flavour bound quoted here, $M_{\rm KK} = 47.26\,{\rm TeV}$ at 50% acceptance for $g_s^\star = 3$, is an independent $\Delta F = 2$ result and is substantially stronger than that external floor.

Gauge-coupling convention. The paper headline convention is $g_s^\star = 3$, corresponding to the no-UV-boundary-term comparison convention used in the Csaki–Falkowski–Weiler discussion. The stored code path uses the perturbative four-dimensional QCD coupling $g_s \simeq 1.05$ at the KK scale as its default normalization, and we quote the perturbative values beside the $g_s^\star = 3$ values so that the rescaling is transparent. The stronger-coupling value $g_s^\star \simeq 6$ appears only when comparing to the CFW boundary-term default, where their published RS marker is 21 TeV rather than the 10.5 TeV no-UV-boundary-term marker. It is not a third headline convention for this paper.

Lepton-sector scope. This repository is the working home for both quark and lepton 5D-flavour studies, but the present methodology note and paper address the quark sector only. The lepton-sector directories `neutrinos/`, `flavorConstraints/`, and `yukawa/` are follow-up work for a separate analysis. They have not been audited, rerun, or reinterpreted under the present `deltaf2.py` updates, which affect the quark $\Delta F = 2$ pipeline.

A.4 Invalidation-gate rerun

The final-claims invalidation gate was cleared only after the hadronic-input sign-off in `docs/phase_logs/phase2_h5_signoff.md` and the Wilson-RG sign-off in `docs/phase_logs/phase2_h6_signoff.md`. The benchmark ε_K ratio shift is the product $6.2388 \times 3.6052 = 22.49$, which is far above the 10% rerun threshold in `docs/paper_execution_decisions.md`. The post-audit rerun report is `docs/phase_logs/invalidation_gate_rerun.md`; it records the nine SLURM job IDs, the final states, the exact RUNA halt-trigger check, and the before/after percentile table. The factor 22.49 is a benchmark ratio, not a constant multiplier for every random draw: in the 3 TeV RUNA tile the median per-draw ε_K ratio shift is $18.55\times$, and the median max ratio shift is $18.11\times$, because the operator mixture and the max-over-systems observable vary draw by draw. The rerun nevertheless makes ε_K overwhelmingly dominant: it binds 99.50% of the post-audit RUNA PDG-passing ensemble.